

Problem Statement – 04

- **Problem Statement Title- AI-Powered Behavioral Anomaly Detection for Cybersecurity**
- **Theme - Cybersecurity**
- **PS Category - Software**
- **Student Name - AYUSHI**
- **Student ID - 23BAI10811**

AI-Powered Behavioral Anomaly Detection for Cybersecurity

Proposed Solution

- Hybrid AI system using **Isolation Forest + LSTM Autoencoder**.
- Detects behavioural anomalies in users, service accounts, and IoT devices.
- Classifies attacks and provides explainable alerts.

How it addresses the problem

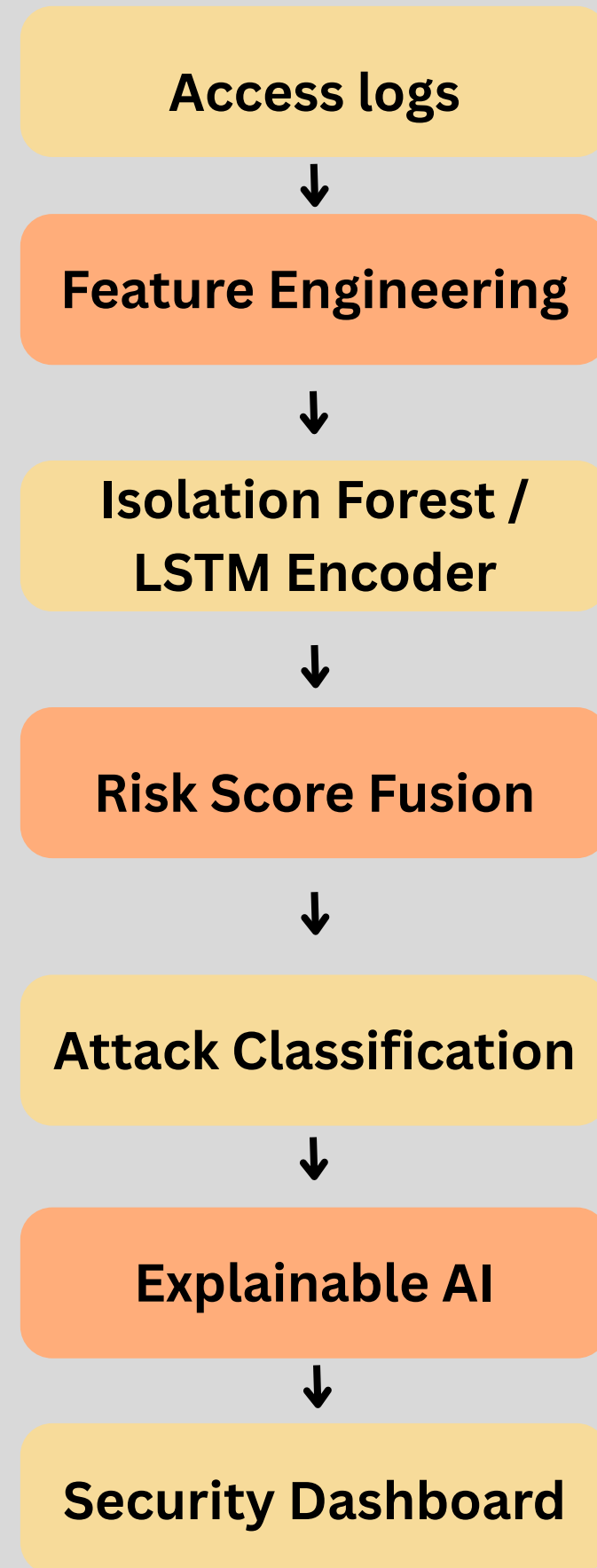
- Learns normal behaviour instead of using fixed rules.
- Detects unknown and insider attacks in real time.
- Reduces false alerts with adaptive learning.

Innovation & Uniqueness

- Hybrid anomaly detection model.
- Handles **cold-start** and **concept drift**.
- Explainable AI with unified risk scoring.

TECHNICAL APPROACH

Flowchart



Python



Pytorch



Tech Stack

FEASIBILITY AND VIABILITY

Feasibility

- Open-source & cost-effective
- Scalable architecture
- Real-world deployment ready

Challenges

- Limited labelled data
- Cold-start detection
- Concept drift

Mitigation Strategies

- Synthetic data generation
- Hybrid AI model
- Adaptive learning (EWMA)

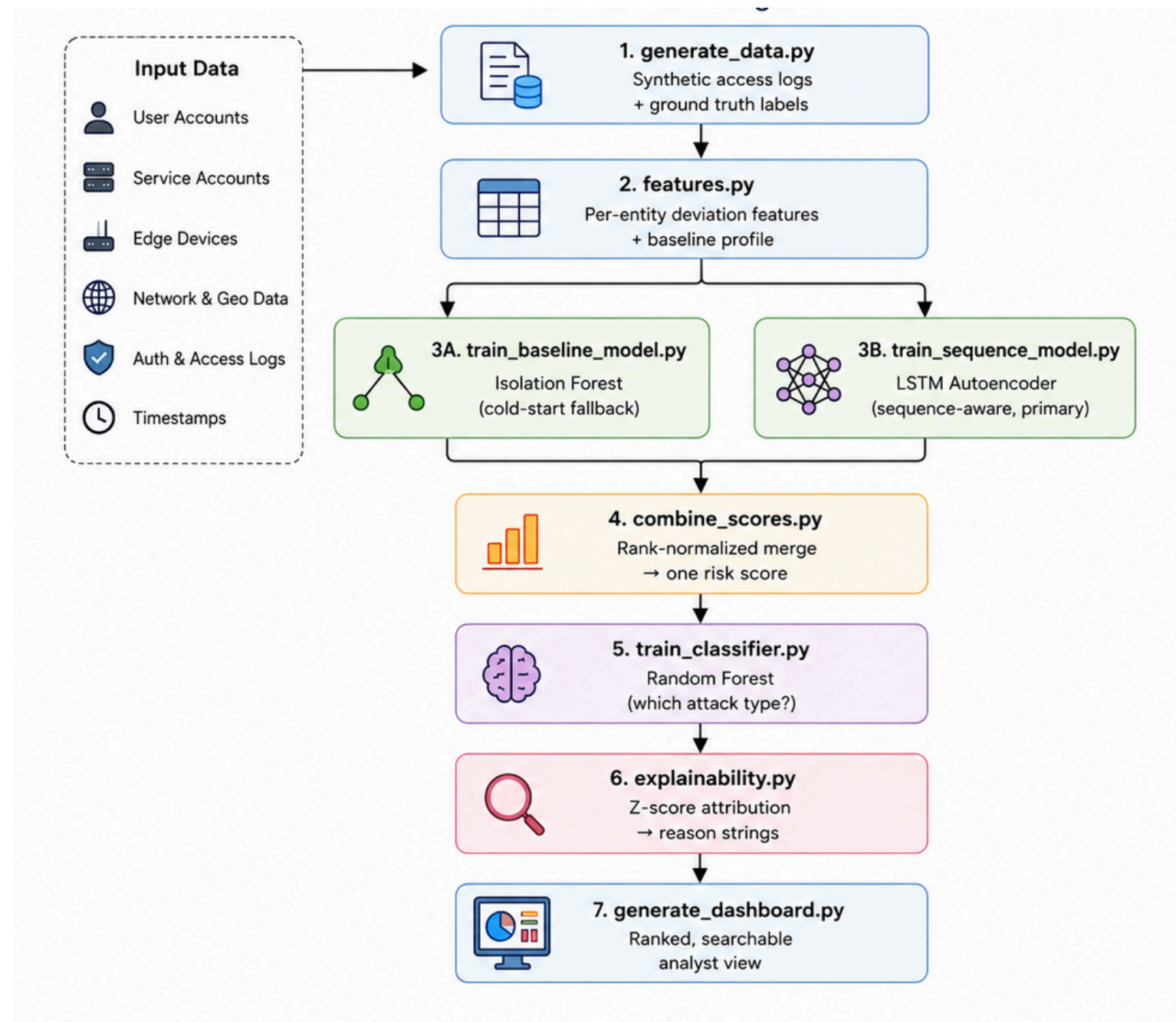
ARTIFACTS

Github Link

<https://github.com/ayushikashyap1207/honeywell2>

Dashboard

Architecture
Diagram



Behavioural anomaly detection - analyst alert queue
Ranked by risk score, within the top 1% alert budget. Click a row for contributing factors and entity history.

18,644 Total sessions scored	186 Alert budget (top 1%)	89.2% Precision within queue	402 True anomalies (all data)	32 Entities monitored
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Search entity id... All predicted types

Rank	Entity	Type	Risk	Detection path	Predicted attack	Confidence	Ground truth
1	user_d9210195	user	1.00	Sequence (LSTM)	brute_force	99%	anomaly
2	user_66ba42e5	user	1.00	Cold-start (Isolation Forest)	low_and_slow_exfiltration	42%	anomaly
3	user_d9210195	user	1.00	Sequence (LSTM)	brute_force	99%	anomaly
4	user_eb94151d	user	1.00	Sequence (LSTM)	brute_force	99%	anomaly
5	user_d9210195	user	1.00	Sequence (LSTM)	brute_force	99%	anomaly
6	user_d9210195	user	1.00	Sequence (LSTM)	brute_force	98%	anomaly
7	user_d9210195	user	1.00	Sequence (LSTM)	brute_force	95%	anomaly
8	user_d9210195	user	1.00	Sequence (LSTM)	brute_force	99%	anomaly

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WHY THIS WAS FLAGGED

Flagged due to: many failed logins from this source IP in a short window + failed authentication attempt(s) + implausible geographic velocity (impossible travel).

TOP CONTRIBUTING FEATURES

ip_failed_5min	z = 13000000
failed_auth_attempts	z = 1000000
geo_km_per_hr	z = 5998.3

RECENT ENTITY HISTORY

2026-07-01 09:32:33.089717	- Frankfurt_DE, resource_051,resource_034, auth=token, dur=380s
2026-06-30 16:44:01.187053	- Moscow_RU, login_endpoint, auth=token, dur=2s
2026-06-30 09:42:41.050872	- Frankfurt_DE, resource_028,resource_113,resource_114, auth=token, dur=452s
2026-06-30 09:34:20.563279	- Frankfurt_DE, resource_059,resource_064,resource_083,resource_041, auth=token, dur=163s
2026-06-30 08:51:39.941363	- Frankfurt_DE, resource_034,resource_118,resource_059, auth=token, dur=35s
2026-06-29 09:54:14.823837	- Frankfurt_DE, resource_092,resource_084,resource_059,resource_118, auth=token, dur=245s
2026-06-29 09:52:58.675410	- Frankfurt_DE, resource_114,resource_059, auth=token, dur=240s
2026-06-29 09:41:01.918532	- Frankfurt_DE, resource_084,resource_059, auth=token, dur=30s

RESEARCH AND REFERENCES

- Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation Forest. ICDM.
- Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory (LSTM).
- Sakurada, M., & Yairi, T. (2014). Autoencoder-based Anomaly Detection.
- scikit-learn Documentation
- PyTorch Documentation